

Spin echo and Nuclear Magnetic Resonance  
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Nuclei in samples of water, glycerine and dilute Ferric Nitrate were subjected to a constant magnetic field of strength  $0.425 \pm 0.001$  T, then were perturbed by a radio-frequency pulse of duration  $100 \pm 1$  ms. The longitudinal relaxation times for these samples were found to be  $28.2 \pm 3.0$  ms,  $299.4 \pm 32.2$  ms and  $263.4 \pm 26.1$  respectively, while the transverse relaxation times were found to be  $15.0 \pm 0.9$  ms,  $116.2 \pm 8.2$  ms and  $114.3 \pm 11.5$  ms. The key findings of this experiment are that transverse relaxation times are shorter than longitudinal relaxation times, and that more viscous materials have reduced relaxation times.

## I. Introduction

I. I. Rabi of Columbia University made the first accurate measurements of nuclear magnetic moments in 1938 using magnetic resonance absorption of molecular beams [1]. He was awarded the Nobel Prize in Physics in 1944 for his work. In 1946, Felix Bloch and Edward Mills Purcell expanded the technique for use on liquids and solids, for which they shared the Nobel Prize in Physics in 1952 [2]. Varian Associates developed the first NMR unit, called NMR HR-30, in 1952 [3]. Since then NMR has been extensively used in medicine in the form of magnetic resonance imaging. NMR is used industrially mainly for routine analysis of chemicals. The technique is also used to measure the ratio between water and fat in foods, monitor the flow of corrosive fluids in pipes, or to study molecular structures such as catalysts. Rabi, Bloch, and Purcell observed that magnetic nuclei, like  $^1\text{H}$  and  $^{31}\text{P}$  can absorb RF energy when placed in a magnetic field. Different atomic nuclei within a molecule resonate at different frequencies for the same magnetic field strength. The observation of these magnetic resonance frequencies allow an observer to discover chemical and structural information about the molecule, meaning NMR can be used to provide detailed information of functional groups, topology, dynamics and the three-dimensional structure of molecules in a sample.

A magnetic moment ( $\mu$ ) will seek to align with an externally applied magnetic field ( $B_0$ ). It will experience a torque ( $\tau$ ) given by the vector

cross product  $\tau = \mu \times B_0$ . The energy ( $E$ ) of the magnetic moment ( $\mu$ ) would be given by the vector dot product:  $E = -\mu \cdot B_0$ . When perfectly aligned parallel to  $B_0$ ,  $\mu$  will be in its lowest energy state and experience no torque. The magnetic moment ( $\mu$ ) of a particle is proportional to its spin ( $I$ ),  $\mu = \gamma I$ , where  $\gamma$  is the gyromagnetic ratio. For a nucleus with spin  $I$  the number of eigenstates is  $2I + 1$ . For the hydrogen nucleus with  $I = \frac{1}{2}$ , there are 2 possible spin states, spin up (parallel) and spin down (anti-parallel). The energy gap ( $\Delta E$ ) between two nuclear spin states is given by the Zeeman equation,  $\Delta E = \gamma \hbar B_0$ . Thus for resonance to occur, the frequency of incident radiation must satisfy,  $\omega = \gamma B_0$ , where  $\omega$  is the Larmor frequency.

When considering macroscopic samples, the population of energy levels will follow the Boltzmann distribution,  $N_+/N_- = \exp[-\Delta E / kT]$ , where  $N_+$  and  $N_-$  represent the number of spins one would expect to measure in the spin-up and spin-down configurations. At room temperature in a field of 0.425 Tesla, this equation predicts that  $N_+$  and  $N_-$  are nearly equal; only a small excess ( $\sim 3 \times 10^{-6}$ ) of spins can be expected to be found in the lower energy (spin-up) state when measured. This small excess leads to a net magnetisation vector, which points in the direction of  $B_0$  at equilibrium. To induce precession,  $M$  must be displaced out of its equilibrium alignment by an external injection of energy. This comes from a second magnetic field (denoted  $B_1$ ) oscillating at the Larmor frequency.  $B_1$  is generated by passing an

alternating current through a nearby radiofrequency (RF) coil. The amount of rotation the net magnetization (M) experiences can be controlled by varying how long the radiofrequency pulse is applied for.

Consider the spin system relative to a set of axes, (x', y', z') rotating at the resonant frequency about the axis of B<sub>0</sub>, which we can label as the z-axis. In this frame the effect of B<sub>0</sub> disappears as there is no precession of M around the z-axis. The B<sub>1</sub> field is no longer rotating but appears static, we can assume it points along the x' axis. Since B<sub>1</sub> is the only magnetic field acting on M in this frame of reference, M will precess around B<sub>1</sub> with a frequency  $f = \gamma$ , meaning we can vary where M points by varying the length of time B<sub>1</sub> is applied for.

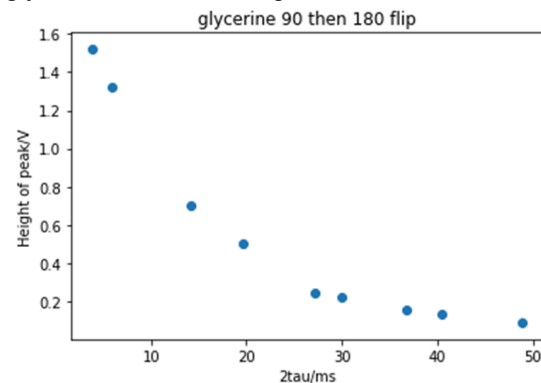
T<sub>1</sub> relaxation is the process by which the net magnetization (M) returns to its initial maximum value (M<sub>0</sub>) parallel to B<sub>0</sub>. As the longitudinal component of M (M<sub>z</sub>) grows toward M<sub>0</sub>, the energy of the spin system decreases as more spins are "favouring" the lower energy (spin-up, parallel) orientation. The energy is transferred into nearby nuclei, atoms, and molecules through collisions, rotations, or electromagnetic interactions. T<sub>2</sub> relaxation is the process by which the transverse components of magnetization (M<sub>xy</sub>) dephase. Anything causing T<sub>1</sub> relaxation also causes T<sub>2</sub> relaxation, but T<sub>2</sub> relaxation can occur without T<sub>1</sub> relaxation. Major causes of T<sub>2</sub>-only relaxation include dephasing by static local field disturbances and flip-flop exchanges between spins. T<sub>2</sub>\* can be considered an effective T<sub>2</sub>. T<sub>2</sub>\* results principally from inhomogeneities in the main magnetic field. These inhomogeneities may be the result of intrinsic defects in the magnet itself or from susceptibility-induced field distortions produced by the tissue or other materials placed within the field.

Assuming the spins are pointing in the z' direction initially, switching B<sub>1</sub> on for a time  $\pi/2\gamma B_1$  will cause the spins to end up pointing in

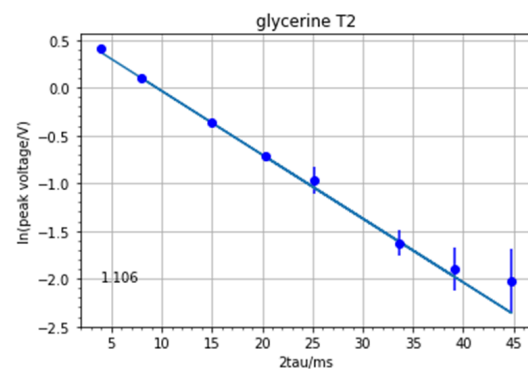
the y' direction. This is defined as a 90 degree pulse, a 180 pulse can be defined in a similar fashion. After a 90 pulse, the spins are all rotating in phase relative to the coil which produced the pulse and so will induce a current in it. This signal is known as free induction and will decay rapidly due to the spins losing phase coherence. After a time t the spins will form a pancake in the x'-y' plane, if a 180 pulse is then applied the spins will refocus along the y' axis after a further time t. The effect of inhomogeneities in the main magnetic field has been made to cancel out, so . This will again induce a signal in the detecting coil, known as the spin echo. The amplitude of the echo will depend on the random loss of phase coherence. The effect of inhomogeneities in the main magnetic field has been made to cancel out, so Assuming T<sub>1</sub> is long, echo height

$$h \propto e^{\frac{-2\tau}{T_2}}$$

The graph of h against tau for al sample of glycerine is shown in figure 1.



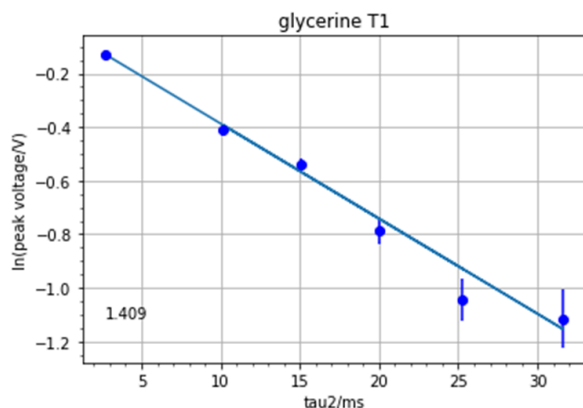
Equation 1 was then modified to give a graph of the form  $y = mx + c$ , the gradient of this graph can be used to give a value for T<sub>2</sub>. The linearised graph can be seen in fig 2.



To determine T1, a second 90 degree pulse was applied a time tau after the first 90 degree pulse. This flips the pancake about the x' axis into the x'z plane. Any change in the angle the spins make with the z' axis now requires a change in energy of the system, which could only come from T1 relaxation. The energy is transferred into nearby nuclei, atoms, and molecules through collisions, rotations, or electromagnetic interactions. After a further time interval tau2, a third 90 pulse was applied, completing the inversion and allowing a spin echo to be formed. The amplitude of the echo now has a dependence on tau2, so

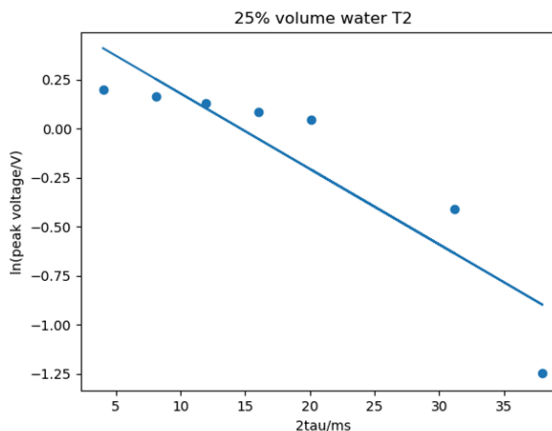
$$h \propto e^{\frac{-\tau_2}{T_1}}$$

When measuring T1 the repetition rate had to be increased to around 10 seconds to get a maximum echo height. We thought this could be due to the spins not fully returning to their equilibrium position of pointing in the z' axis. The linearised graph used to find T1 for a sample of glycerine is shown in figure 3.



A major source of error in the experiment came from diffusion of the nuclei being measured. This became a major factor with longer tau. Figure 4 shows how diffusion changes the general trend of the graph for longer tau. To counteract this effect only shorter tau readings were used to calculate the gradient. A systematic source of error came from the purity of the samples being used. With the equipment available it could not be guaranteed that the glycerine samples did

not contain water, or that the water used did not contain impurities.



When calculating the error on individual readings the precision of the instruments used was taken into account. The width of the spin echo and the fluctuation in the amplitude were used to assign an error to each reading. A line of best fit was drawn by minimising the square of the residuals, this was used to calculate the final uncertainties on the relaxation times.

The longitudinal relaxation times T1 for glycerol, water and dilute Ferric Nitrate were found to be  $28.2 \pm 3.0$  ms,  $299.4 \pm 32.2$  ms and  $263.4 \pm 26.1$  respectively, while the transverse relaxation times T2 were found to be  $15.0 \pm 0.9$  ms,  $116.2 \pm 8.2$  ms and  $114.3 \pm 11.5$  ms. The key findings of this experiment are that transverse relaxation times are shorter than longitudinal relaxation times, and that more viscous materials have reduced relaxation times. The addition of Ferric nitrate, a paramagnetic salt, reduced relaxation times, having a greater effect on T1.

In conclusion,

#### References:

1. Rabi, I.I.; Zacharias, J.R.; Millman, S. & Kusch, P. (1938). "A New Method of Measuring Nuclear Magnetic Moment". Physical Review
2. Filler, Aaron (2009). "The History, Development and Impact of Computed Imaging in Neurological Diagnosis and

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